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This document outlines and breaks down the scoring algorithm used by Salud.

1. Overview

Algorithm Name: Salud Health Score

Purpose:

This algorithm is used to assess the healthiness of an individual food item by providing a score from 0-1. It takes into account user preferences and is designed to be tunable.

Scope:

This algorithm is used after a food item is scanned and its raw data is stored.

Constraints:

1. Does not know the exact quantity of each ingredient
2. Ingredient quality is dependent on a predetermined score (can change over time) and some ingredients may not be accounted for

2. High-Level Description

Conceptual Summary:

This algorithm outputs a score that is made up of the 4 weighted scores based on the following areas:

1. Energy Ratio
2. Macro Ratio
3. Ingredient Quality
4. Nutrient Density

Each of these sub scores are weighted depending on their impact on health. These weights can be tweaked over time to improve the accuracy of the algorithm.

1. Energy Ratio Score:

This sub score is based on the caloric density of a given food item. This score is then weighted based on the user's goals. (ex. A user who is trying to lose weight's score will be more heavily impacted by a calorically dense food)

2. Macro Ratio Score

This sub score is based on how well a given food item's macro ratio matches the user's macro goals.

3. Ingredient Quality Score:

This sub score is based on the following factors: ingredient risk factor & ingredient quantity. Ingredient risk factor is a predetermined score based on the current research. Ingredient quantity is the position in the ingredient list. The exact ingredient quantity can not be determined without reverse engineering a food which is unfeasible and unreliable.

4. Nutrient Density Score:

This sub score reflects how nutritious a food is. This score can be tuned depending on how much of an emphasis on that density.

Inputs and Outputs:

Type	Name	Description	Format/Example
Input	Calories	Number of calories per 100g of food	c
Input	Caloric Density	Calories per gram of food	d_c
Input	Energy Score Constant	Decimal from 0-1 that controls how steep score fall off is (Note: this weight can vary depending on a user's goals)	k_E
Input	Energy Ratio Sub Score	Decimal from 0-1 that represents a good or bad energy score	s_E
Input	Grams of Nutrients	Number of grams of nutrients per 100g of the food item	n
Input	Nutrient Density	How much nutrition per calorie of food	d_n
Input	Nutrient Density Constant	Decimal from 0-1 that controls the emphasis placed on nutritional density	k_N
Input	Nutrient Density Sub Score	Decimal from 0-1 that represents a good or bad nutrient density score	s_N
Input	Risk Score	A score from 0-10 that assess how risky the consumption of that ingredient is (Note: this score is pulled from a database)	r_i
Input	Normalized Risk Score	A conversion of the initial Risk Score normalized to a decimal from 0-1	$\hat{r}_i$
Input	Ingredient Weight	A weight assigned to an ingredient based on its position in the ingredient list	w_i

Input	Ingredient Quality Constant	A constant that can be used to tune the score. It is used in the calculation of w_i .	k_q
Input	Ingredient Quality Sub Score	A decimal from 0-1 that represents a good or bad ingredient quality score	s_q
Input	User Carb Percent Goal	A decimal that represents the goal percentage of carbs in the user's diet	g_c
Input	User Protein Percent Goal	A decimal that represents the goal percentage of protein in the user's diet	g_p
Input	User Fat Percent Goal	A decimal that represents the goal percentage of fat in the user's diet	g_f
Input	Food Carb Percentage	A decimal that represents the percentage of a food item that is carbs	f_c
Input	Food Protein Percentage	A decimal that represents the percentage of a food item that is protein	f_p
Input	Food Fat Percentage	A decimal that represents the percentage of a food item that is fat	f_f
Input	Macro Ratio Difference	A decimal from 0-2 that represents how close a food item's macro ratio and a user's goal macro ratio match. 0 represents a perfect match and 2 represents the worst possible match	D
Input	Macro Ratio Sub Score	A decimal from 0-1 that represents how close a user's goal macro ratio matches.	s_M
Input	Energy Score Weight	A decimal from 0-1 that represents how much emphasis on energy the final score represents	w_E
Input	Macro Score Weight	A decimal from 0-1 that represents how much emphasis on macro ratio matching the final score represents	w_M
Input	Ingredient Quality Score Weight	A decimal from 0-1 that represents how much emphasis on ingredient quality the final score represents	w_Q
Input	Nutritional Density Score Weight	A decimal from 0-1 that represents how much emphasis on nutritional density the final score represents	w_N

Output	Overall Health Score	A decimal from 0-1 that represents how healthy a food item is. It comprises the above 4 sub scores.	S _H
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3. Algorithm Steps

This algorithm is broken into 5 parts

1. Compute Energy Ratio Sub Score
2. Compute Macro Ratio Sub Score
3. Compute Ingredient Quality Sub Score
4. Compute Nutritional Density Sub Score
5. Compute Overall Health Score

3.1 Energy Ratio Sub Score

$$d_c = \frac{c}{100}$$

$$s_E = \frac{1}{1+kd_c}$$

3.2 Macro Ratio Sub Score

$$D = |f_c - g_c| + |f_p - g_p| + |f_f - g_f|, \quad \text{where } g_c + g_p + g_f = 1 \quad \& \quad f_c + f_p + f_f = 1$$

$$s_M = 1 - \frac{D}{2}$$

3.3 Ingredient Quality Sub Score

$$\hat{r}_i = \frac{r_i}{10}$$

$$w_i = k_q^{i-1}, \quad \text{where } 0 < k_q < 1$$

$$s_Q = \frac{\sum_{i=1}^n w_i \hat{r}_i}{\sum_{i=1}^n w_i}$$

3.4 Nutritional Density Sub Score

$$d_n = \frac{n}{c}$$

$$S_N = \frac{1}{1+k_q d_n}$$

3.5 Overall Health Score

$$S_H = S_E^{w_E} \cdot S_M^{w_M} \cdot S_Q^{w_Q} \cdot S_N^{w_N}$$